SOLVE-IT Exercise: Weakness Enumeration with TRWM

Overview for instructors

Version	0.2 DRAFT
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Prerequisites	● An introduction to the SOLVE-IT knowledge base ● An introduction to the TRWM methodology
Intended Learning Outcomes (ILOs)	At the end of this exercise you will be able to: ● Use a systematic approach to determine the technique results, weaknesses and mitigations for a technique in the SOLVE-IT knowledge base.
Instructor notes	This exercise will ask students to take a technique from SOLVE-IT, or one that is missing, and to systematically enumerate the weaknesses and mitigations associated with the technique. TRWM is the general approach of systematically considering: Technique, Result, Weaknesses and Mitigations. TRWM-A is the use of TRWM where specifically the ASTM E3016-18 error classifications are used to iterate through potential weaknesses. It also introduced the *SOLVE-IT TRWM Workbook,* which takes the form of a Google Sheet that can be duplicated and used by students to work through the systematic approach.

SOLVE-IT Exercise: Weakness Enumeration with TRWM

1. Pre-requisites

To complete this exercise you should have a general understanding of the SOLVE-IT knowledge base. You should also have been introduced to the TRWM[1] methodology, and have access to the TRWM Helper Workbooks[2] to help with this.

2. Aim

The aim of this exercise is for you to systematically enumerate the weaknesses for a technique in the SOLVE-IT knowledge base.

3. Overview of the Tasks

This exercise consists of a workthrough of the a Google Worksheet, which uses the TRWM-A[3] methodology. This section consists of 'Getting started' followed by subsections for each stage of the walkthrough.

3.1 Getting started

Open the template TRWM worksheet in Google Drive[4]. You should see something similar to what is shown in Figure 1 below:

[1] TRWM is the approach that systematically reviews in turn Technique, technique Result, Weaknesses, and Mitigations.
[2] Previous versions of these sheets were referred to as SOLVE-IT Helper for Weakness and Mitigation Analysis (SHWAMA) but this is no longer used to simplify the terminology used.
[3] TRWM-A is the version of the TRWM methodology that specifically uses the ASTM E3016-18 error classifications to iterate through potential weaknesses.
[4] Examples here use worksheet version 2.41, but the latest version of the worksheet is linked from this page: https://github.com/SOLVE-IT-DF/solve-it-education/tree/main/trwm-explained

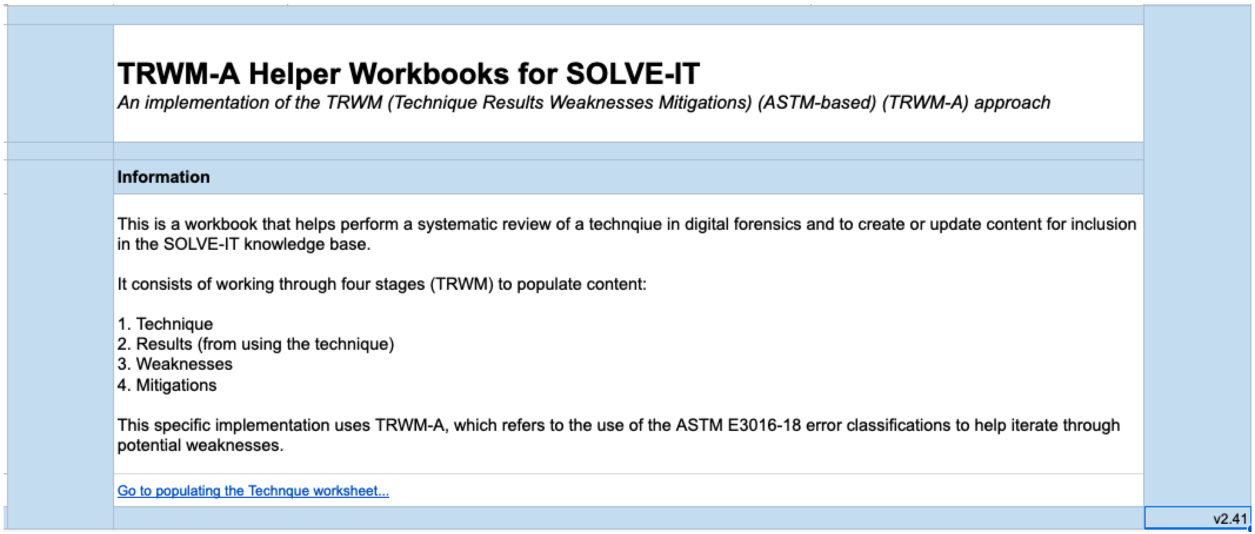

Figure 1: Shows the 'Information' worksheet of the TRWM Helper Workbook in Google Drive.

This workbook will be a read only copy. You should create a copy that you can edit, as shown below in Figure 2:

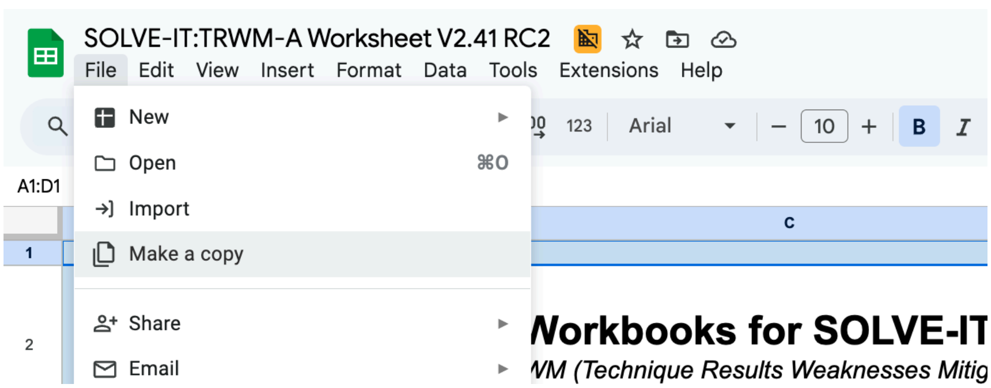

Figure 2: Shows the interface for creating a copy of the read only workbook.

An overview of the process is shown in the information sheet - TRWM refers to: Technique, Results, Weaknesses, Mitigations. These steps are discussed in the subsections that follow.

3.2 (T) - Technique

First, the technique must be selected. You may be asked by your instructor to consider a specific technique or you may be free to choose your own.

If these are not details provided already, establishing the definition(s) of what this technique entails is the starting point. Look at digital forensic literature and propose a definition for this technique.

You may at this point also want to look for other details to complete, e.g. synonyms, examples of tools and processes that make use of this technique, and/or cases that have used it.

These details will help you with scoping during the next parts of the task.

Within the helper worksheet, you will find placeholders to assist you with completing these fields. An example for a completed technique (T1073: Calendar app examination) is shown in Figure 3 below. This example is used throughout the rest of the exercise explanation[5].

[5] The full worksheet for the *T1073:Calendar app examination* example is available here: https://docs.google.com/spreadsheets/d/1z2akrdahCnWDK7O0zER5WCUi4Tpsh3D94hGjvR7OB2E

TRWM-A Helper Workbooks for SOLVE-IT

An implementation of the TRWM (Technique Results Weaknesses Mitigations) (ASTM-based) (TRWM-A) approach

Technique stage: Explanation

This phase is to document the nature of the technique, and provide a description and examples of use.

Ideally the name and description of the technique would come from an academic, government or industry source, but some techqniues do not have a clean definition in literature.

Additional explanatory notes for each field are provided on the right hand side.

Field	***Contents***	***Notes***
SOLVE-IT repo link:	https://github.com/SOLVE-IT-DF/solve-it/blob/main/data/techniques/T1073.json	*You can copy the link from the project repo address bar e.g. https://github.com/SOLVE-IT-DF/solve-it/blob/main/data/techniques/T1000.json*
Technique ID:	T1073	*This is only needed if this is an existing techqniue, ignore this if a new technique is being proposed.*
Technique name:	Calendar app examination	*Current or proposed name for the technique.*
Technique description:	The analysis of built in or third-party calendar applications.	*A description or defintion of the technique. This will ideally be backed by literature from academia, government or industry.*
Synonyms:	online calendar examination, agenda app examination, scheduling app examination	*Many techniques in digital forensics are referred to by multiple names. Synonyms for the technique can be added here to assist SOLVE-IT users in finding the relevant technique.*
Technique details:	This technique has the potential to associate the owner of the smartphone to specific locations at specific times (adapted from Alghafli et al. 2011)	*This field can be used to provide more details beyond the definition, to keep that definition concise.*
Examples:	Google Calendar, Microsoft Outlook, Apple Calendar (iCal), FAntastical, Thunderbird	*Here examples can be provided that make use of the technique.*
References:	Alghafli, K.A., Jones, A. and Martin, T.A., 2011. Guidelines for the digital forensic processing of smartphones. 9th Australian Digital Forensics Conference, Edith Cowan University, Perth Western Australia, 5th -7th December 2011 Chu, H.C., Wang, G.G. and Park, J.H., 2015. The digital fingerprinting analysis concerning google calendar under ubiquitous mobile computing era. Symmetry, 7(2), pp.383-394. Metz (2009) E-mail and appointment falsification analysis, Forensic Focus, https://www.forensicfocus.com/articles/e-mail-and-appointment-falsification-analysis	*Where possible, please provide references to support the details provided above.* *References should not be added just because they are about a topic, but should have meaningful implications in terms of explaining a technique.* *For large references, consider supplying the page or chapter number if appropriate.* *Harvard referencing is used for now, but we will move to bibtex representations in future.*

Go to populating the Results worksheet...

Figure 3: Shows the 'Technique' worksheet of the workbook in Google Drive.

3.3 (R) - Results

This next step is more involved and requires consideration of the digital forensic technique and the nature of the results of using the technique. There are multiple approaches to assist with this:

- **If the technique is the examination of a type of app, you should consider that app carefully.**
    - There may be multiple implementations of the app type, e.g. for *T1072: Chat app examination*, examples are WhatsApp, Telegram, Signal, and iMessage, but there are many others. The more implementations you consider the more comprehensive your output will be.
    - You should explore an example of the app under consideration. Think about what pieces of data are forensically useful. This might include user generated content, timestamps, settings and configurations.

- **Examine forensic tool implementation outputs**
    - The existing data extraction performed by forensic tools of the app or apps of the same type can also be insightful.
    - Review the tool output for processing the app or a similar app and note other potentially useful forensic artefacts.
    - Other techniques used by tools can also be considered e.g. timeline generation. Think about what results these techniques produce.
- **Compile a list of all potential Digital Forensic Tool Results (DFTRs)**
    - Using the artefacts identified and other information collected, compile a list of all the potential results from using the technique.
    - Within the helper sheet, there are spaces for a name for the result, a description, and you can also consult the CASE Ontology documentation class A-Z[6] to see if any of the types of results are already included in the ontology. If any result is not, you can also note a potential new CASE class in the column provided.

The relevant parts of the helper sheet are shown in Figure 4 below populated for the Calendar example. You can see some CASE classes already exist and several others are proposed.

While linking with CASE is useful, as it checking what can be modelled that is related, but you should primarily focus on what makes the most sense as DFTR concepts. CASE representation can come at a later date if needed.

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Results stage: Explanation

This phase is to identify the results that can emerge from using the specifed digital forensic technique.

These are referred to as: Digital Forensic Technique Results/Outputs (DFTRs). They will be used in the next stage to help enumerate potential weaknesses of using a digital forensic technique.

These can be identified in multiple ways.

* If the techqniue involves the extraction of artefacts, e.g. *photos app examination,* then it is useful to experiment with one or more of these types of app and note the data points that could be relevant to a forensic investigation, e.g. photo content, photo taken time, album names, list of photos in album.

* It is also useful to look at forensic tools that already process this type of app and record the type of artefacts that are already extracted, e.g. latitude and longitude recorded in a photo.

Record each of the potential results below. You can provide a short name, a description, and also consult the CASE Ontology a-z list of classes (linked below) to see if this type of result is already included in the ontology. If it is not, you can note a potential new CASE class in the 'Potential CASE class' column.

The worksheet *2a_Results_Notes* can be used to document this process for tranparancy as to how these DFTRs were determined.

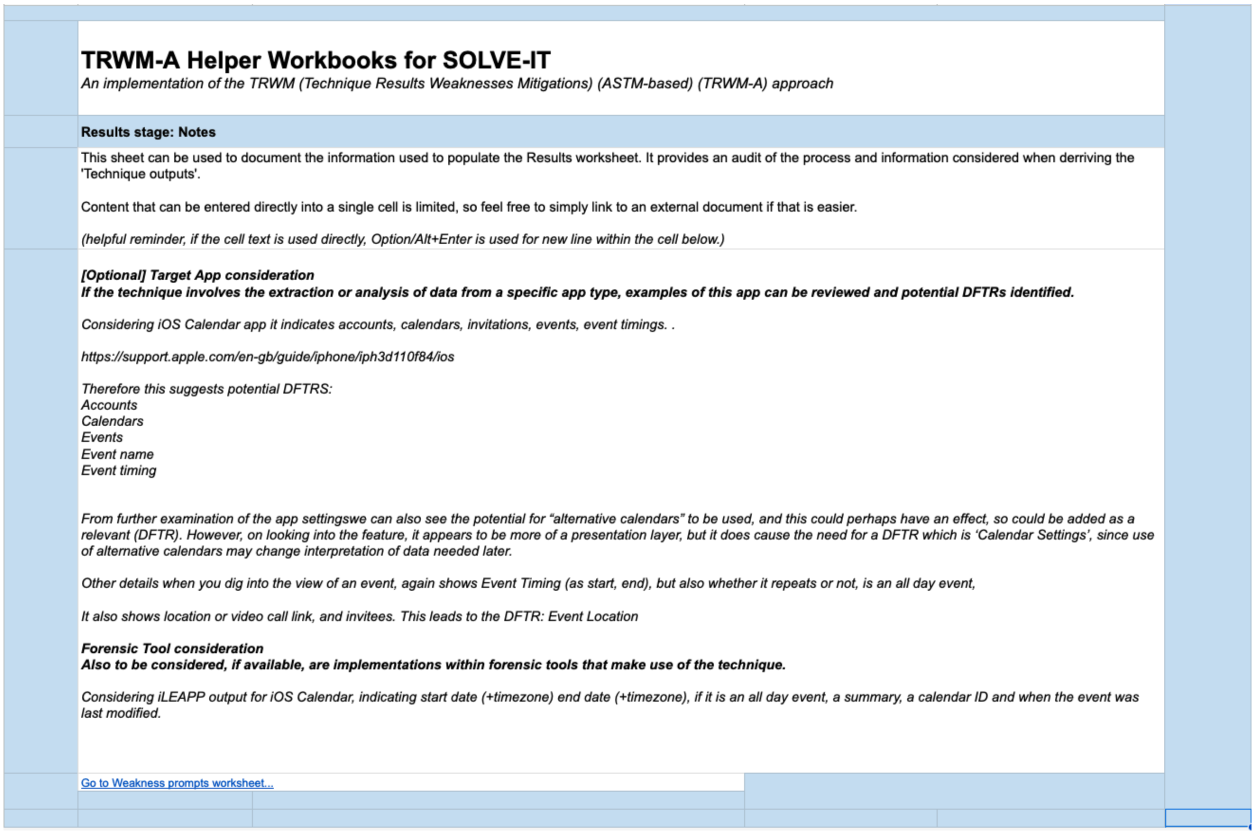
DFTR ID	Name	Description	CASE class (refer to a-z list)	Potential CASE class
R1	Calendar accounts	Accounts that contain individual calendars	observable:Account	
R2	Calendars	A calendar that contains calendar entries (events)	observable:Calendar	
R3	Event	An entry in a calendar referring to a specific event at a time	observable:CalendarEntry	
R4	Event name	The name of the event		CalendarEntryName
R5	Event timing	The timings of the event, start, end, duration, repeat status, 'all day' status		CalendarEventDateTime
R6	Event location	The location of the event (physical, online meeting)	observable:location, observable:URL	OnlineMeetingID, OnlineMeetingPassword
R7	Event invitees	The identifiers of people invited to the event	identity:Identity	
R8	Calendar settings	Settings associated with the calendar app (e.g. if alternative calendars are in use)		
R9				
R10				
R11				
R12				

Go to Weakness prompts worksheet...

Figure 4: Shows the 'Results' worksheet of the workbook in Google Drive.

[6] https://ontology.caseontology.org/documentation/entities-az.html

Also within newer versions of the worksheet is the worksheet **2a_Results_Notes** which provides a space to record notes as to how the DFTRs were derived. Since content in an individual cell is limited, a link to an external document can also be provided.



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Results stage: Notes

This sheet can be used to document the information used to populate the Results worksheet. It provides an audit of the process and information considered when derriving the 'Technique outputs'.

Content that can be entered directly into a single cell is limited, so feel free to simply link to an external document if that is easier.

(helpful reminder, if the cell text is used directly, Option/Alt+Enter is used for new line within the cell below.)

[Optional] Target App consideration
If the technique involves the extraction or analysis of data from a specific app type, examples of this app can be reviewed and potential DFTRs identified.

Considering iOS Calendar app it indicates accounts, calendars, invitations, events, event timings. .

https://support.apple.com/en-gb/guide/iphone/iph3d110f84/ios

Therefore this suggests potential DFTRS:
Accounts
Calendars
Events
Event name
Event timing

From further examination of the app settingswe can also see the potential for "alternative calendars" to be used, and this could perhaps have an effect, so could be added as a relevant (DFTR). However, on looking into the feature, it appears to be more of a presentation layer, but it does cause the need for a DFTR which is 'Calendar Settings', since use of alternative calendars may change interpretation of data needed later.

Other details when you dig into the view of an event, again shows Event Timing (as start, end), but also whether it repeats or not, is an all day event,

It also shows location or video call link, and invitees. This leads to the DFTR: Event Location

Forensic Tool consideration
Also to be considered, if available, are implementations within forensic tools that make use of the technique.

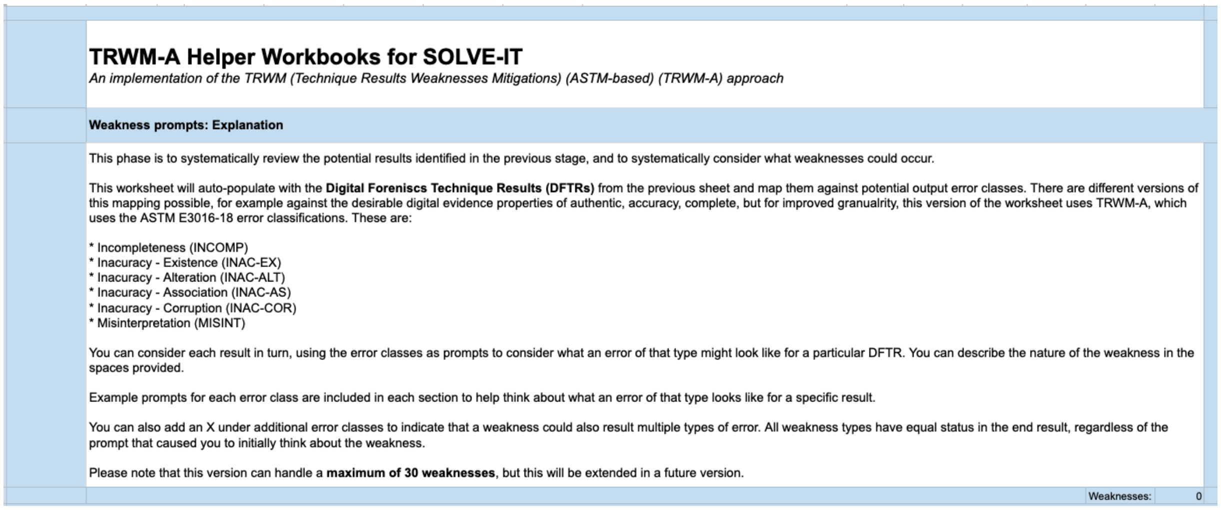
Considering iLEAPP output for iOS Calendar, indicating start date (+timezone) end date (+timezone), if it is an all day event, a summary, a calendar ID and when the event was last modified.

Go to Weakness prompts worksheet...

Figure 4a: Shows the 'Results Notes' worksheet of the workbook in Google Drive.

3.4 (W) - Weaknesses

This stage is focused on enumerating the weaknesses associated with the technique. The guidance provided in the helper sheet is shown in Figure 5.



TRWM-A Helper Workbooks for SOLVE-IT

An implementation of the TRWM (Technique Results Weaknesses Mitigations) (ASTM-based) (TRWM-A) approach

Weakness prompts: Explanation

This phase is to systematically review the potential results identified in the previous stage, and to systematically consider what weaknesses could occur.

This worksheet will auto-populate with the **Digital Foreniscs Technique Results (DFTRs)** from the previous sheet and map them against potential output error classes. There are different versions of this mapping possible, for example against the desirable digital evidence properties of authentic, accuracy, complete, but for improved granualrity, this version of the worksheet uses TRWM-A, which uses the ASTM E3016-18 error classifications. These are:

* Incompleteness (INCOMP)
* Inacuracy - Existence (INAC-EX)
* Inacuracy - Alteration (INAC-ALT)
* Inacuracy - Association (INAC-AS)
* Inacuracy - Corruption (INAC-COR)
* Misinterpretation (MISINT)

You can consider each result in turn, using the error classes as prompts to consider what an error of that type might look like for a particular DFTR. You can describe the nature of the weakness in the spaces provided.

Example prompts for each error class are included in each section to help think about what an error of that type looks like for a specific result.

You can also add an X under additional error classes to indicate that a weakness could also result multiple types of error. All weakness types have equal status in the end result, regardless of the prompt that caused you to initially think about the weakness.

Please note that this version can handle a **maximum of 30 weaknesses**, but this will be extended in a future version.

Weaknesses: 0

Figure 5: Shows the introductory text of the Weakness Prompts worksheet.

Within the helper sheet, the Digital Forensic Tool Results (DFTRs) are auto populated from the previous sheet. Figure 6 shows the section for one of those DFTRs (the Calendar Accounts). Within each result, there are spaces provided, along with prompts, to consider what an error looks like (based on the ASTM E3016-18 error classes) for each of the DFTRs.

This allows each result type to be systematically considered against the possible error classes resulting in a thorough assessment of the weaknesses associated with a digital forensic technique.

Note that not all error types will have entries for all DFTRs.

Also note that on the right, there are prefilled checkmarks showing the primary error class for a weakness (based on the subsection that has been completed). You may also override these with additional error classes if appropriate. For example, For *calendar accounts*, the weakness “Presentation of an account that does not exist” has a main category of INAC-EX, but as this could also result in misinterpretation, the MISINT box has been manually checked.

			INCOMP	INAC-EX	INAC-ALT	INAC-AS	INAC-COR	MISINT	Error class summary
R1	**Calendar accounts**	Accounts that contain individual calendars	*Note, these are precompleted to align with the main error class, but additional error classes sometimes apply to a weakness and can be marked with an X.*						
INCOMP	***Prompts: failure to recover live artefacts, failure to recover deleted artefacts, other reasons why an artefact might be missed?***								
	Failure to recover live calendar accounts		X						INCOMP
	Failure to recover deleted but recoverable calendar accounts		X						INCOMP
			X						INCOMP
			X						INCOMP
			X						INCOMP
INAC-EX	***Prompts: presenting an artefact for something that does not exist***								
	Presentation of an account that does not exist			X				X	INAC-EX, MISINT
				X					INAC-EX
				X					INAC-EX
				X					INAC-EX
				X					INAC-EX
INAC-ALT	***Prompts: modifying the content of some digital data***								
					X				INAC-ALT
					X				INAC-ALT
					X				INAC-ALT
					X				INAC-ALT
					X				INAC-ALT
INAC-AS	***Prompts: presenting live data as deleted and vice versa***								
	Presenting a live account as deleted					X			INAC-AS
	Presenting a deleted account as live					X			INAC-AS
						X			INAC-AS
						X			INAC-AS
						X			INAC-AS
INAC-COR	***Prompts: could the process corrupt data, could the process fail to detect corrupt data?***								
							X		INAC-COR
							X		INAC-COR
							X		INAC-COR
							X		INAC-COR
							X		INAC-COR
MISINT	***Prompts: could results be presented in a way that encourages misinterpretation?***								
	Failing to display that an account was flagged as inactive							X	MISINT
								X	MISINT
								X	MISINT
								X	MISINT
								X	MISINT

Figure 6: Shows the spaces for weakness enumeration of a single Digital Forensic Tool Result (DFTR) (*Calendar accounts* in this case) mapped against the ASTM E3016-18 error classifications.

Once this sheet is complete, there is a summary provided in the next worksheet (*3b_Weakness (aggregate) (ASTM)*). This is shown below in Figure 7 and provides a more compact overview of the details that have been provided. This is useful to review the weaknesses in isolation and ensure that they will ‘stand alone’ when entered into the knowledge base. For example "message not recovered" may be better phrased as "chat message not recovered" to ensure context is preserved in the weakness name when viewed in isolation.

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Weakness aggregation: Explanation

There is nothing to edit on this sheet currently. It presents a summary of the weaknesses identtified and their error classes.

This is useful to review to ensure that the weakness name/description 'stands alone' e.g. "message not recovered" may be better as "chat message not recovered" to ensure context is preserved in the weakness name.

Data on this sheet is used to help automatically populate other sheets.

Automatic Aggregated Weakness List	**ASTM**						
	INCOMP	**INAC-EX**	**INAC-ALT**	**INAC-AS**	**INAC-COR**	**MISINT**	**Error class summary**
Failure to recover live calendar accounts	X						INCOMP
Failure to recover deleted but recoverable calendar accounts	X						INCOMP
Presentation of a calendar account that does not exist		X					INAC-EX
Presenting a live calendar account as deleted				X			INAC-AS
Presenting a deleted calendar account as live				X			INAC-AS
Failing to display that a calendar account was flagged as inactive						X	MISINT
Failure to recover live calendars	X						INCOMP
Failure to recover deleted but recoverable calendars	X						INCOMP
Presenting a calendar that does not exist		X					INAC-EX
Presenting a live calendar as deleted				X			INAC-AS
Presenting a deleted calendar as live				X			INAC-AS
Failure to recover deleted but recoverable calendar events	X						INCOMP
Failure to recover calendar events removed as archived or synced online	X						INCOMP
Failure to recover live calendar events from a calendar	X						INCOMP
Presenting a calendar entry that does not exist		X					INAC-EX
Associating a calendar event with the wrong calendar				X			INAC-AS
Associating a calendar event with the wrong account				X			INAC-AS
Failure to detect modification to the calendar event					X		INAC-COR
Failure to recover a calendar event name	X						INCOMP
Presenting a calendar event name that does not exist		X					INAC-EX
Missing calendar events due to not taking into account event recurrences	X						INCOMP
Presenting a calendar event at an incorrect time due to incorrect timezone parsing			X				INAC-ALT
Failure to recover calendar event location information	X						INCOMP
Presenting a location not present in the calendar event		X					INAC-EX
Failure to recover calendar event attendee information	X						INCOMP
Failure to recover calendar event attendee acceptance detail	X						INCOMP
Presenting an invitee not assoicated with the calendar event		X					INAC-EX
Failure to display invitee calendar event acceptance detail						X	MISINT
Failure to display alternative calendars in forensic tool, when they were in use by the target calendar app						X	MISINT
	-	-	-	-	-	-	-

Figure 7: Once complete, the weaknesses will be summarized in the “3b_Weakness (aggregate (ASTM)” worksheet.

3.5 (M) - Mitigations

This final stage involves considering the mitigations for each of the weaknesses identified. Figure 8 shows the introductory text for the next sheet (4_Mitigations).

The worksheet will auto populate the weaknesses and their error class as a reminder.

While an approach as systematic as the weakness enumeration does not yet exist, there is some guidance provided as to the types of mitigations that can exist. However, they are not exhaustive and digital forensic literature, tools, techniques, and your own ideas and experience should be drawn upon to populate the mitigations for a particular weakness. You can also make use of existing mitigations in the SOLVE-IT knowledge base.

Existing mitigations will appear in a drop down menu (see Figure 8a), but it is highly likely you will need to add new mitigations which can be entered as free text.

Also please note that the mitigation list has to be manually updated from SOLVE-IT, so please check the last updated date to be aware of the last manually mitigation list synchronisation. The mitigation list is available as a hidden sheet in the workbook.

An example subset of mitigations for the calendar app examination example are shown in Figure 9.

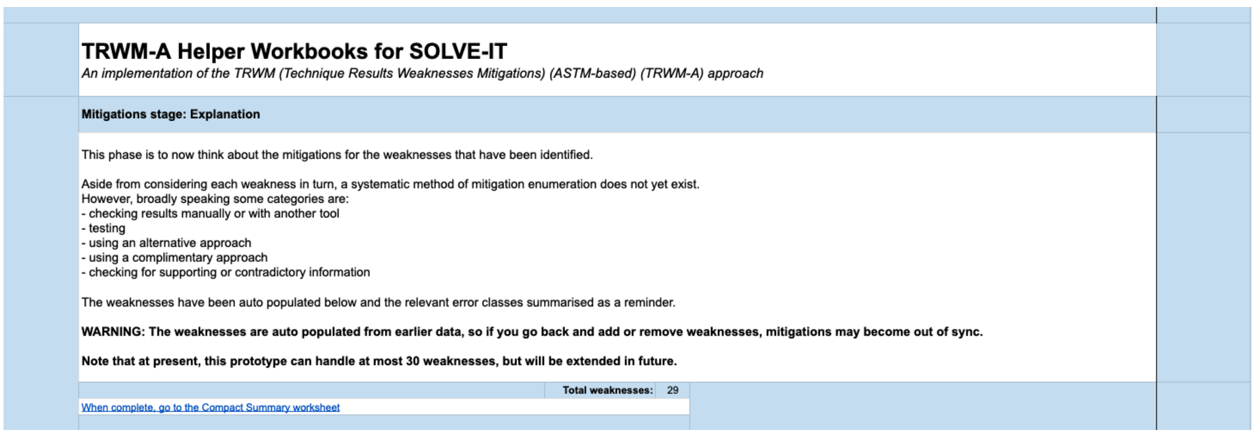

Figure 8: The introductory text from the "4_Mitigations" worksheet.

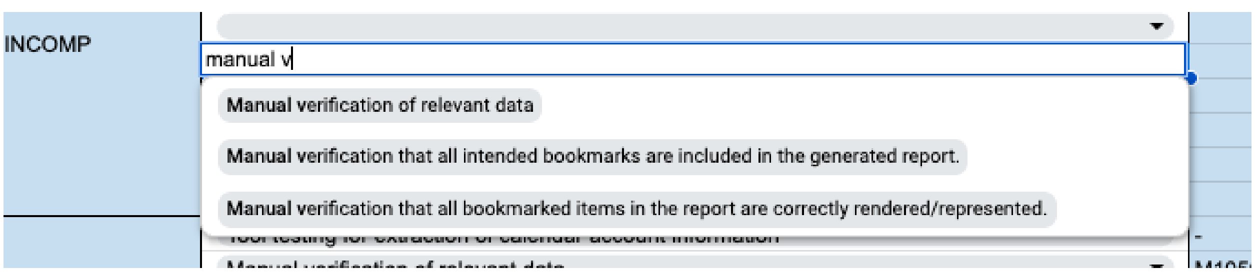

Figure 8a: Shows the dropdown menu for existing mitigations in SOLVE-IT. Manual mitigations can also be entered as free text.

Weaknesses (coresponding results are deliberately not repeated here to ensure that weakness description make sense as stand alone descriptions)	**Reminder of error classes of weakness**	**Proposed mitigation descriptions** (dropdowns are provided for existing mitigations, but **new ones can be added in place of the dropdown**) **NOTE: mitigation list last updated: 2025-08-18**	**Existing mitigation ID if applicable** (autopopulated)
Failure to recover live calendar accounts	INCOMP	Tool testing for extraction of calendar account information	-
		Manual verification of relevant data	M1050
		Dual tool verification	M1027
Failure to recover deleted but recoverable calendar accounts	INCOMP	Tool testing for extraction of calendar account information	-
		Manual verification of relevant data	M1050
		Dual tool verification	M1027
Presentation of a calendar account that does not exist	INAC-EX	Tool testing for extraction of calendar account information	-
		Manual verification of relevant data	M1050
		Dual tool verification	M1027
Presenting a live calendar account as deleted	INAC-AS	Tool testing for extraction of calendar account information	-
		Manual verification of relevant data	M1050
		Dual tool verification	M1027

Figure 9: Shows the section of the worksheet where mitigations can be proposed.

Once complete. The summary of mitigations proposed are provided in the worksheet ‘4b_Mitigation Summary’. This can be helpful in identifying duplicates or close duplicates and can be used to refine the list. Figure 10 shows the results from the calendar example.

In terms of some high-level guidance, it is not an exact science as to where to be specific and where to generalise. In some cases generic ‘Manual verification of relevant data’ is used, and in some it is more specific e.g. “Manually check the calendar account status (live/deleted)”.

However, typically the tool testing based mitigations are useful to specify what needs to be tested as these can form the basis of tangible tool testing programmes.

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Explanation

There is nothing to edit on this sheet currently. It simply presents a deduplicated summary of the mitigations proposed.

This can be useful to help identify if individual mitigations could be merged.

Mitigation	Occurances
Compare calendar event creation and modification times	1
Dual tool verification	26
Ensure 'active status' of calendar accounts is viewed	1
Ensure alternative calendars are presented	1
Ensure calendar event invitee acceptance status is presented	1
Ensure that deleted calendar events are searched for	1
Manual check of calendar account status	1
Manual verification of relevant data	25
Recover data from online account and extract calendar entries	1
Recover old copies or backups of calendar data and extract calendar entries	1
Tool testing for extraction of calendar account information	5
Tool testing for extraction of calendar event names	2
Tool testing for extraction of calendar events	3
Tool testing for extraction of calendar events and account association	1
Tool testing for extraction of calendar events and calendar association	1
Tool testing for extraction of calendar events with alternative calendars	1
Tool testing for extraction of calendar events with attendees	3
Tool testing for extraction of calendar events with locations	2
Tool testing for extraction of calendar events with reccurrences	1
Tool testing for extraction of calendar events with time zones	1
Tool testing for extraction of calendars	5

Figure 10: Shows a summary of the mitigations proposed and the number of occurrences.

3.6 Combination Summary Sheet

After this process, the ‘Compact Summary’ sheet in the workbook will summarize this information in a single table. An example is shown below in Figure 11 and this sheet can used as the basis for populating a technique, its weaknesses and mitigations within the SOLVE-IT knowledge base.

Generated using TRWM/SHWAMA	*v2.2*					
Technique name:	Calendar app examination					
Technqiue description:	The analysis of built in or third-party calendar applications.					
Synonyms:						
Details:	This techqniue has the potential to associate the owner of the smartphone to specific locations at specific times (adapted from Alghafli et al. 2011)					
Examples:						
CASE output classes:	observable:Account, observable:Calendar, observable:CalendarEntry, observable:location, observable:URL, identity:Identity					
Propsed CASE classes:	CalendarEntryName, CalendarEventDateTime, OnlineMeetingID, OnlineMeetingPassword					
References:	Alghafli, K.A., Jones, A. and Martin, T.A., 2011. Guidelines for the digital forensic processing of smartphones. 9th Australian Digital Forensics Conference, Edith Cowan University, Perth Western Australia, 5th -7th December 2011 Metz (2009) E-mail and appointment falsification analysis, Forensic Focus, https://www.forensicfocus.com/articles/e-mail-and-appointment-falsification-analysis					
Weaknesses and mitigations:		**M1**	**M2**	**M3**	**M4**	**M5**
Failure to recover live calendar accounts	INCOMP	*Tool testing for extraction of calendar account information*	*Manual verification of relevant data*	*Dual tool verification*		
Failure to recover deleted but recoverable calendar accounts	INCOMP	*Tool testing for extraction of calendar account information*	*Manual verification of relevant data*	*Dual tool verification*		
Presentation of an account that does not exist	INAC-EX, MISINT	*Tool testing for extraction of calendar account information*	*Manual verification of relevant data*	*Dual tool verification*		
Presenting a live account as deleted	INAC-AS	*Tool testing for extraction of calendar account information*	*Manual verification of relevant data*	*Dual tool verification*		
Presenting a deleted account as live	INAC-AS	*Tool testing for extraction of calendar account information*	*Manual verification of relevant data*	*Dual tool verification*		
Failing to display that an account was flagged as inactive	MISINT	*Ensure 'active status' of calendar accounts is viewed*	*Manual check of calendar account status*			
Failure to recover live calendars	INCOMP	*Tool testing for extraction of calendars*	*Manual verification of relevant data*	*Dual tool verification*		
Failure to recover deleted but recoverable calendars	INCOMP	*Tool testing for extraction of calendars*	*Manual verification of relevant data*	*Dual tool verification*		
Presenting a calendar that does not exist	INAC-EX	*Tool testing for extraction of calendars*	*Manual verification of relevant data*	*Dual tool verification*		

Figure 11: Shows an extract from the final summary sheet generated, including the technique details, weakness list, and mapped mitigations.

3.7 TSV Summary Sheets

Three additional sheets are available: _tsv_technique, _tsv_weaknesses, and _tsv_mitigations (shown in Figure 12, 13 and 14). These can be downloaded as TSV files, as shown in Figure 15. It will shortly be possible to use these TSV files to auto-generate JSON files to be added directly into the SOLVE-IT knowledge base. See updates to the SOLVE-IT scripts for the latest updates.

Technique ID:	T1073
Technique name:	Calendar app examination
Technique description:	The analysis of built in or third-party calendar applications.
Synonyms:	online calendar examination, agenda app examination, scheduling app examination
Technique details:	This technique has the potential to associate the owner of the smartphone to specific locations at specific times (adapted from Alghafli et al. 2011)
Examples:	Google Calendar, Microsoft Outlook, Apple Calendar (iCal), FAntastical, Thunderbird
References:	Alghafli, K.A., Jones, A. and Martin, T.A., 2011. Guidelines for the digital forensic processing of smartphones. 9th Australian Digital Forensics Conference, Edith Cowan University, Perth Western Australia, 5th -7th December 2011 Chu, H.C., Wang, G.G. and Park, J.H., 2015. The digital fingerprinting analysis concerning google calendar under ubiquitous mobile computing era. Symmetry, 7(2), pp.383-394. Metz (2009) E-mail and appointment falsification analysis, Forensic Focus, https://www.forensicfocus.com/articles/e-mail-and-appointment-falsification-analysis
CASE Classes	observable:Account, observable:Calendar, observable:CalendarEntry, observable:location, observable:URL, identity:Identity

Figure 12: Shows the tsv_technique sheet

Weakness ID	Weakness	INCOMP	INAC-EX	INAC-ALT	INAC-AS	INAC-COR	MISINT	Summary	Mitgation 1	Mitigation 2
	Failure to recover live calendar accounts	X						INCOMP	Tool testing for extraction of calendar a	Manual verifica
	Failure to recover deleted but recoverable calendar accounts	X						INCOMP	Tool testing for extraction of calendar a	Manual verifica
	Presentation of a calendar account that does not exist		X					INAC-EX	Tool testing for extraction of calendar a	Manual verifica
	Presenting a live calendar account as deleted				X			INAC-AS	Tool testing for extraction of calendar a	Manual verifica
	Presenting a deleted calendar account as live				X			INAC-AS	Tool testing for extraction of calendar a	Manual verifica
	Failing to display that a calendar account was flagged as inactive						X	MISINT	Ensure 'active status' of calendar accou	Manual check
	Failure to recover live calendars	X						INCOMP	Tool testing for extraction of calendars	Manual verifica
	Failure to recover deleted but recoverable calendars	X						INCOMP	Tool testing for extraction of calendars	Manual verifica
	Presenting a calendar that does not exist		X					INAC-EX	Tool testing for extraction of calendars	Manual verifica
	Presenting a live calendar as deleted				X			INAC-AS	Tool testing for extraction of calendars	Manual verifica
	Presenting a deleted calendar as live				X			INAC-AS	Tool testing for extraction of calendars	Manual verifica
	Failure to recover deleted but recoverable calendar events	X						INCOMP	Tool testing for extraction of calendar e	Dual tool verific
	Failure to recover calendar events removed as archived or synced online	X						INCOMP	Recover data from online account and	Recover old co
	Failure to recover live calendar events from a calendar	X						INCOMP	Tool testing for extraction of calendar e	Dual tool verific
	Presenting a calendar entry that does not exist		X					INAC-EX	Tool testing for extraction of calendar e	Dual tool verific
	Associating a calendar event with the wrong calendar				X			INAC-AS	Tool testing for extraction of calendar e	Dual tool verific
	Associating a calendar event with the wrong account				X			INAC-AS	Tool testing for extraction of calendar e	Dual tool verific
	Failure to detect modification to the calendar event					X		INAC-COR	Compare calendar event creation and r	
	Failure to recover a calendar event name	X						INCOMP	Tool testing for extraction of calendar e	Dual tool verific
	Presenting a calendar event name that does not exist		X					INAC-EX	Tool testing for extraction of calendar e	Dual tool verific
	Missing calendar events due to not taking into account event recurrences	X						INCOMP	Tool testing for extraction of calendar e	Dual tool verific
	Presenting a calendar event at an incorrect time due to incorrect timezone parsing			X				INAC-ALT	Tool testing for extraction of calendar e	Dual tool verific
	Failure to recover calendar event location information	X						INCOMP	Tool testing for extraction of calendar e	Dual tool verific
	Presenting a location not present in the calendar event		X					INAC-EX	Tool testing for extraction of calendar e	Dual tool verific
	Failure to recover calendar event attendee information	X						INCOMP	Tool testing for extraction of calendar e	Dual tool verific
	Failure to recover calendar event attendee acceptance detail	X						INCOMP	Tool testing for extraction of calendar e	Dual tool verific
	Presenting an invitee not assoicated with the calendar event		X					INAC-EX	Tool testing for extraction of calendar e	Dual tool verific
	Failure to display invitee calendar event acceptance detail						X	MISINT	Ensure calendar event invitee acceptar	Dual tool verific
	Failure to display alternative calendars in forensic tool, when they were in use by the target calendar app						X	MISINT	Tool testing for extraction of calendar e	Dual tool verific

Figure 13: Shows the tsv_weaknesses sheet

ID	Mitigation
todo	Compare calendar event creation and modification times
M1027	Dual tool verification
todo	Ensure 'active status' of calendar accounts is viewed
todo	Ensure alternative calendars are presented
todo	Ensure calendar event invitee acceptance status is presented
todo	Ensure that deleted calendar events are searched for
todo	Manual check of calendar account status
M1050	Manual verification of relevant data
todo	Recover data from online account and extract calendar entries
todo	Recover old copies or backups of calendar data and extract calendar entries
todo	Tool testing for extraction of calendar account information
todo	Tool testing for extraction of calendar event names
todo	Tool testing for extraction of calendar events
todo	Tool testing for extraction of calendar events and account association
todo	Tool testing for extraction of calendar events and calendar association
todo	Tool testing for extraction of calendar events with alternative calendars
todo	Tool testing for extraction of calendar events with attendees
todo	Tool testing for extraction of calendar events with locations
todo	Tool testing for extraction of calendar events with reccurrences
todo	Tool testing for extraction of calendar events with time zones
todo	Tool testing for extraction of calendars

Figure 14: Shows the tsv_mitigations sheet

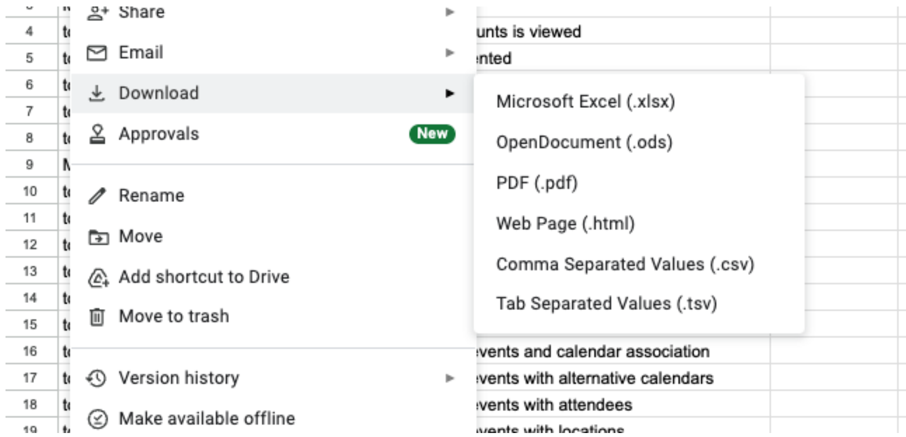

Figure 15: Shows the 'Download as Tab Separated Values (.tsv) option'.